AnalystLab Africa

Week 2: University Academic Management Analytics & Student Success Intelligence Project

Business Insights Report

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1. Total number of students

	count bigint
1	10000

2. Student enrolment by faculty

	faculty_name character varying (250)	total_enrollments bigint
1	Faculty of Business & Economics	6467
2	Faculty of Applied Sciences	6381
3	Faculty of Computing & Informatics	6348
4	Faculty of Education & Social Scien...	6333
5	Faculty of Engineering	6163
6	Faculty of Health Sciences	6144

3. Student enrolment by department

	department_name character varying (250)	total_enrollments bigint
1	Economics	2729
2	Artificial Intelligence	2623
3	Data Science	2619
4	Statistics	2593
5	Information Technology	2577
6	Accounting	2566
7	Software Engineering	2554
8	Public Health	2526
9	Computer Science	2494
10	Electrical Engineering	2438
11	Education	2416
12	Mechanical Engineering	2363
13	Nursing	2357
14	Finance	2333
15	Psychology	2290

4. Student enrolment by programme

	programme_name character varying (250)	total_enrollments bigint
1	MSc Computer Science	2133
2	BSc Computer Science	2122
3	MSc AI	2098
4	BCom Finance	2088
5	MSc Software Engineer...	2087
6	BSc Data Science	2083
7	MSc IT	2062
8	BEng Mechanical Engin...	2052
9	MEng Electrical	2048
10	BCom Accounting	2043
11	BEng Electrical Enginee...	2037
12	MSc Data Science	1999
13	BSc Nursing	1995
14	BSc Psychology	1987
15	BSc Statistics	1971
16	BEd Education	1966
17	BSc Information Techn...	1954
18	MA Economics	1938
19	MBA Finance	1935
20	BSc Economics	1919
21	MEng Mechanical	1904
22	BSc Public Health	1899
23	BSc Software Engineeri...	1874
24	BSc Artificial Intelligence	1869
25	MBA Accounting	1867

5. Student enrolment by campus

	campus_name character varying (250)	total_enrollments bigint
1	Nakuru Campus	9703
2	Mombasa Campus	9918
3	Nairobi Main Campus	10053
4	Kisumu Campus	10149
5	Eldoret Campus	10107

6. Gender distribution of students

	gender character varying (100)	total_students bigint	percentage numeric
1	Female	5005	50.1
2	Male	4995	50.0

7. Enrolment trends by academic year

	academic_year_id character varying (25)	total_students bigint
1	AY2022	3417
2	AY2023	3306
3	AY2024	3277

8. Most popular programmes

	programme_name character varying (250)	total_students bigint
1	MSc Computer Science	430
2	BSc Computer Science	422
3	MSc AI	419
4	BCom Finance	418
5	MSc Software Engineering	417
6	BSc Data Science	416
7	BCom Accounting	414
8	MEng Electrical	412
9	BEng Electrical Engineering	411
10	BEng Mechanical Engineeri...	409

9. Most popular courses.

	course_code integer	course_title character varying (250)	total_enrollments bigint
1	13402	Networks Education	236
2	1102	Machine Learning Science	227
3	3101	Machine Learning Science	226
4	6302	Principles of Accounting	203
5	5302	Introduction Engineering	203
6	8202	Machine Learning Economi...	192
7	6201	Machine Learning Accounti...	192
8	11301	AI Nursing	190
9	8302	Theory of Economics	187
10	14401	Introduction Psychology	125

10. Attendance rate by course
https://drive.google.com/file/d/1OXwLex5wjIAgkpM61SMYP70tunf02dro/view?usp=share_link

11. Attendance rate by lecturer
https://drive.google.com/file/d/1xSPxTUDUYCyYBKX3KX6Dwi2W4pxJgkOF/view?usp=share_link

12. Attendance rate by department
https://drive.google.com/file/d/18BRh-72gMW0Jyxl1sW_IS7D0Ah0XUum5/view?usp=share_link

13. Students with attendance below 75%
    https://drive.google.com/file/d/1IvccG1VjoGYvinv2lBw6lUH6R10dYIcZ/view?usp=share_link

14. Average assessment score by course
    https://drive.google.com/file/d/1OXwLex5wjIAgkpM61SMYP70tunf02dro/view?usp=share_link

15. Pass rate by faculty

	faculty_name character varying (250)	exam_results bigint	passed bigint	pass_rate_pct numeric
1	Faculty of Computing & Informatics	6338	5058	79.8
2	Faculty of Engineering	6196	4939	79.7
3	Faculty of Health Sciences	6113	4864	79.6
4	Faculty of Education & Social Scien…	6276	4971	79.2
5	Faculty of Business & Economics	6496	5123	78.9
6	Faculty of Applied Sciences	6378	5025	78.8

16. Pass rate by department

	department_name character varying (250)	exam_results bigint	passed bigint	pass_rate_pct numeric
1	Accounting	2564	2089	81.5
2	Electrical Engineering	2453	1973	80.4
3	Mechanical Engineering	2368	1899	80.2
4	Data Science	2594	2064	79.6
5	Finance	2339	1861	79.6
6	Software Engineering	2556	2031	79.5
7	Artificial Intelligence	2606	2069	79.4
8	Education	2394	1898	79.3
9	Nursing	2382	1886	79.2
10	Public Health	2552	2013	78.9
11	Economics	2717	2140	78.8
12	Psychology	2268	1788	78.8
13	Statistics	2579	2027	78.6
14	Information Technology	2585	2024	78.3
15	Computer Science	2515	1965	78.1

17. Pass rate by course

	programme_name character varying (250)	exam_results bigint	passed bigint	pass_rate_pct numeric
1	BSc Economics	1888	1540	81.6
2	MA Economics	1992	1607	80.7
3	MBA Accounting	1852	1492	80.6
4	MSc Computer Science	2149	1725	80.3
5	BSc Nursing	2007	1608	80.1
6	MEng Mechanical	1893	1513	79.9
7	MBA Finance	1942	1547	79.7
8	MSc IT	2022	1611	79.7
9	BSc Data Science	2079	1657	79.7
10	BEng Electrical Enginee…	2052	1627	79.3
11	BEng Mechanical Engin…	2043	1621	79.3
12	BSc Artificial Intelligence	1862	1475	79.2
13	BSc Psychology	2016	1597	79.2
14	BCom Finance	2088	1654	79.2
15	BSc Computer Science	2105	1665	79.1
16	BEd Education	1969	1558	79.1
17	BSc Statistics	1973	1561	79.1
18	BSc Information Techn…	1954	1546	79.1
19	MSc Data Science	2001	1578	78.9
20	MEng Electrical	2058	1623	78.9
21	MSc Software Engineer…	2083	1640	78.7
22	BSc Software Engineeri…	1882	1476	78.4
23	BSc Public Health	1862	1458	78.3
24	MSc AI	2091	1636	78.2

20	MEng Electrical	2058	1623	78.9
21	MSc Software Engineer…	2083	1640	78.7
22	BSc Software Engineeri…	1882	1476	78.4
23	BSc Public Health	1862	1458	78.3
24	MSc AI	2091	1636	78.2

| 25 | BCom Accounting | 2067 | 1600 | 77.4 |

18. Top-performing students

	student_id [PK] character varying (50)	first_name character varying (100)	last_name character varying (100)	exams_written bigint	avg_exam_score numeric
1	STU05432	Esther	Adebayo	5	56.4
2	STU05369	Faith	Okeke	5	55.6
3	STU09952	Peter	Ali	4	55.3
4	STU08008	Joy	Abebe	5	55.0
5	STU08374	Daniel	Okafor	5	54.6
6	STU06247	Mary	Bello	5	54.4
7	STU09223	Faith	Mensah	5	54.4
8	STU01399	Esther	Okafor	5	54.0
9	STU08949	James	Diallo	5	53.8
10	STU09046	Samuel	Mwangi	5	53.8

19. Bottom-performing students

	student_id [PK] character varying (50)	first_name character varying (100)	last_name character varying (100)	exams_written bigint	avg_exam_score numeric
1	STU01896	Daniel	Ndlovu	5	12.4
2	STU05454	Michael	Ndlovu	5	13.0
3	STU02220	Joy	Kamau	5	14.4
4	STU02791	Michael	Mensah	5	15.2
5	STU08878	Ruth	Abebe	5	15.4
6	STU08604	Esther	Ndlovu	5	15.8
7	STU08175	Samuel	Mwangi	5	15.8
8	STU08939	Daniel	Adebayo	5	16.0
9	STU07302	Linda	Kone	5	16.0
10	STU06755	Mary	Abebe	5	16.0

20. Lecturer teaching workload
https://drive.google.com/file/d/1-iT5LR8b3xKXj2l_d7-anfivxFgtLNLy/view?usp=share_link

21. Courses with the highest failure rates

	course_code integer	course_title character varying (250)	exam_results bigint	failed bigint	failure_rate_pct numeric
1	6402	Fundamentals Account…	109	34	31.2
2	11201	Advanced Nursing	98	29	29.6
3	9102	Programming Mechani…	103	30	29.1
4	2101	Principles of IT	102	29	28.4
5	10102	Theory of Engineering	103	29	28.2
6	14402	Applied Psychology	96	27	28.1
7	8102	Statistics Economics	93	26	28.0
8	15202	Networks Statistics	104	29	27.9
9	1301	Advanced Science	119	33	27.7
10	5401	Introduction Engineering	116	32	27.6

22. Average GPA by programme

	programme_name character varying (250)	avg_gpa numeric
1	MSc Computer Science	35.55
2	BSc Nursing	35.47
3	MA Economics	35.32
4	MSc IT	35.29
5	MBA Finance	35.25
6	BSc Economics	35.15
7	MBA Accounting	35.08
8	BEng Electrical Enginee…	35.03
9	MSc Software Engineer…	34.98
10	BSc Psychology	34.96
11	MSc Data Science	34.95
12	BSc Public Health	34.94
13	BEng Mechanical Engin…	34.92
14	BSc Artificial Intelligence	34.90
15	BSc Information Techn…	34.89
16	BSc Computer Science	34.82
17	MEng Mechanical	34.80
18	BSc Statistics	34.79
19	BCom Finance	34.77
20	MEng Electrical	34.72
21	BEd Education	34.69
22	MSc AI	34.65
23	BSc Data Science	34.51
24	BSc Software Engineeri…	34.48
25	BCom Accounting	34.44

23. Graduation rate by faculty

	faculty_name character varying (250)	total_students bigint	graduates bigint	graduation_rate_pct numeric
1	Faculty of Applied Sciences	1277	444	34.8
2	Faculty of Education & Social Scien…	1257	435	34.6
3	Faculty of Health Sciences	1225	420	34.3
4	Faculty of Business & Economics	1301	445	34.2
5	Faculty of Engineering	1242	418	33.7
6	Faculty of Computing & Informatics	1269	426	33.6

24. Scholarship distribution across programmes
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25. Executive summary of academic performance

	total_students bigint	total_courses bigint	total_faculties bigint	total_departments bigint	attendance_rate_pct numeric	graduation_rate_pct numeric
1	10000	240	6	15	95.9	34.2